STA6166 Statistical Methods I

Final Project: Multiple Linear Regression and Variable Selection

Total: 100 Points

Student Name: _____ Date: _____

Project Overview

You will analyze a dataset containing one continuous response variable Y and 15 candidate predictors X_1, …, X_{15}. The data were generated from a multiple linear regression model in which exactly 5 predictors have nonzero population regression coefficients; the other 10 have coefficients equal to zero. The identities of the 5 true predictors will not be provided.

Your tasks are to fit the full regression model, examine its assumptions, select predictors using one classical model-selection approach, fit a LASSO regression with a cross-validated tuning parameter, and compare the resulting models. Complete the analysis in R. Unless otherwise stated, use a 5% significance level.

Part 1. Full Multiple Linear Regression and Diagnostics (20 Points)

Question 1. Fit the full model (5 points). Fit the model $Y_i = \beta_0 + \sum_{j=1}^{15} \beta_j X_j + \epsilon_i$ Report the coefficient estimates, standard errors, t-statistics, and p-values. Include the model's R^2 and adjusted R^2.

Fill in the following table using the R output from the full model with 15 predictors.

Coefficient	Point estimate	Standard error	t-statistic	p-value
β_1	1.9115	0.7175	2.664	0.009253
β_2	0.2025	0.6982	0.290	0.772483
β_3	-0.3949	0.6771	-0.583	0.561335
β_4	-2.6473	0.8055	-3.287	0.001481
β_5	1.1445	0.7498	1.526	0.130681
β_6	0.9065	0.7347	1.234	0.220698
β_7	1.2767	0.6619	1.929	0.057121
β_8	0.2487	0.7648	0.325	0.745818
β_9	-0.1041	0.6638	-0.157	0.875731
β_{10}	7.2157	0.7176	10.056	4.49e-16

β_{11}	-0.1946	0.6258	-0.311	0.756631
β_{12}	0.3334	0.6508	0.512	0.609722
β_{13}	-2.6675	0.6812	-3.916	0.000183
β_{14}	-1.6059	0.6871	-2.337	0.021801
β_{15}	0.5023	0.6974	0.720	0.473400

Model fit (full model, R output): Multiple $R^2 = 0.6506$ and adjusted $R^2 = 0.5882$. Residual standard error = 6.427 on 84 degrees of freedom; overall F-statistic = 10.43 on 15 and 84 DF, p-value = 1.442e-13. The intercept estimate is 1.5842 (SE 0.7122, t = 2.224, p = 0.028813).

At the 5% level, X1, X4, X10, X13 and X14 have significant coefficients; X7 is borderline (p = 0.057121). The other nine predictors are not significant.

Question 2. Unequal error variances (5 points). Apply the Breusch–Pagan test to the full regression model.

(a) State the null and alternative hypothesis.

H_0: the error variance is constant, $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2$ for all i (homoscedasticity).

H_1: the error variance depends on the predictors $X_1, \ldots, X_{15}$ (heteroscedasticity).

(b) Write down the test statistic and p-value you get from the R function.

Breusch–Pagan statistic BP = 17.673 with df = 15; p-value = 0.2802.

```
    studentized Breusch-Pagan test

data:  full_model
BP = 17.673, df = 15, p-value = 0.2802
```

(c) State if you have enough evidence to reject H_0.

No. Since p = 0.2802 > 0.05, we fail to reject H_0. There is not enough evidence of unequal error variances; the constant-variance assumption is reasonable for the full model.

Question 3. Normality diagnostic (5 points). Apply the Shapiro–Wilk test to test the normality of residuals of the full regression model.

(a) State the null and alternative hypothesis.

H_0: the regression errors are normally distributed.

H_1: the regression errors are not normally distributed.

(b) Write down the test statistic and p-value you get from the R function.

Shapiro–Wilk statistic W = 0.99144; p-value = 0.7795.

```
    Shapiro-Wilk normality test

data:  res
W = 0.99144, p-value = 0.7795
```

(c) State if you have enough evidence to reject H_0.

No. Since p = 0.7795 > 0.05, we fail to reject H_0; the residuals show no evidence of departure from normality. (The test is applied to OLS residuals, which are slightly dependent and have unequal variances, so the p-value is a descriptive diagnostic rather than an exact test.)

Question 4. Diagnostic plots (5 points). Provide (a) a residuals-versus-fitted plot and (b) a normal Q–Q plot. Briefly discuss whether the plots are consistent with constant error variance and normality.

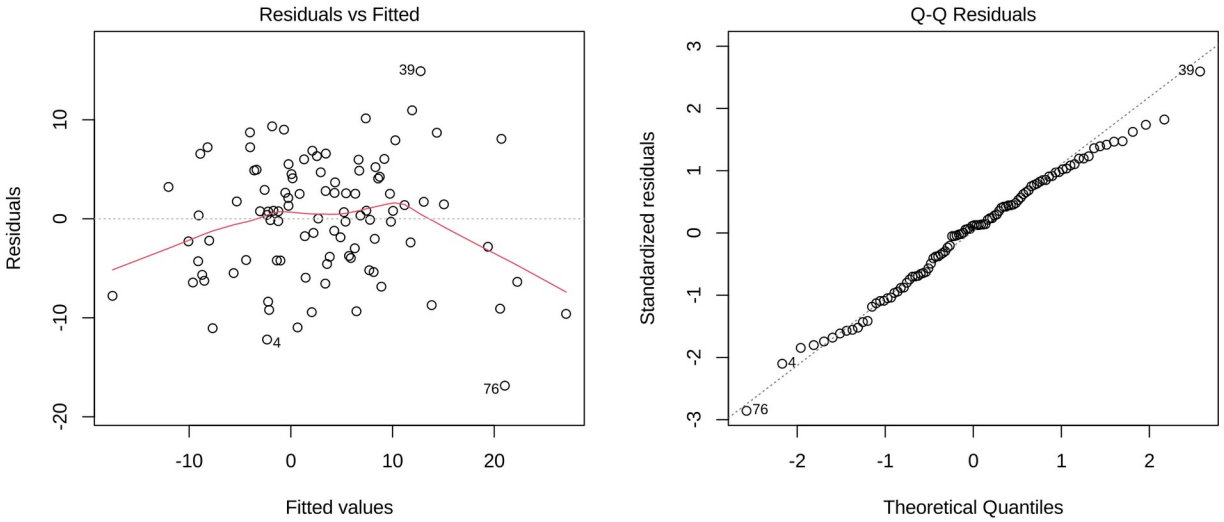


Figure 1. (a) Residuals versus fitted values and (b) normal Q–Q plot of the full model.

(a) Residuals vs fitted: the residuals scatter randomly around zero with roughly the same vertical spread across the range of fitted values; there is no funnel shape. The red smoother bends downward at both ends, but only because of a few points at extreme fitted values (e.g., observation 76), so this is not a systematic curvature. The plot is consistent with constant error variance, in agreement with the Breusch–Pagan test.

(b) Normal Q–Q: the standardized residuals lie close to the reference line, with only small deviations in the tails (observations 39, 76 and 4 are labelled). This is consistent with normal errors, in agreement with the Shapiro–Wilk result. Overall, the full model's assumptions appear reasonable.

Part 2. Classical Variable Selection (30 Points)

Question 5. Choose a search procedure (5 points). Choose ONE procedure: forward selection, backward elimination, or bidirectional stepwise selection. Explain how the procedure (the ONE procedure you choose) adds or removes variables.

Chosen procedure: bidirectional stepwise selection.

The search starts from the intercept-only model Y ~ 1. At each step it evaluates every one-variable move: adding each predictor not yet in the model, and removing each predictor already in the model. It makes the single move that lowers the selection criterion the most. Because removals are allowed at every step, a variable added early can be dropped later if other variables make it redundant (unlike pure forward selection). The search stops when no single addition or removal improves the criterion. In R this is step(null_model, scope = list(lower = Y ~ 1, upper = Y ~ X1 + … + X15), direction = "both", k = log(N)).

Question 6. Choose one selection criterion (5 points). Choose exactly one criterion: AIC, BIC, Mallows' C_p, or adjusted R^2. Explain what the criterion measures and why you chose it. Once selected, keep this criterion fixed throughout Part 2; do not switch criteria after seeing the results.

For AIC or BIC, you may use stats::step() with forward, backward, or bidirectional search. For C_p or adjusted R^2, you may use leaps::regsubsets() with forward or backward search. The seqrep method in regsubsets() is sequential replacement, not bidirectional stepwise regression. If you choose an unsupported combination, you must implement and document the search algorithm yourself. You may revise your initial choice of search procedure before running the analysis to use one of the supported combinations.

Chosen criterion: BIC (Bayesian Information Criterion), minimized.

BIC = $-2 \log L + p \cdot \log(N)$, where L is the maximized likelihood and p is the number of estimated parameters. For a Gaussian linear model this is $N \cdot \log(RSS/N) + p \cdot \log(N)$ plus a constant. The first term measures lack of fit; the second penalizes model complexity, and a smaller BIC indicates a better fit–complexity trade-off.

Reason for choosing it: with N = 100 the penalty is $\log(100) \approx 4.61$ per parameter, much heavier than AIC's 2, so BIC favours parsimonious models. BIC is also consistent: if the true model is among the candidates, BIC selects it with probability approaching 1 as N grows. The project's goal is to recover a sparse set of truly nonzero predictors (exactly 5 of 15), so a criterion aimed at identifying the true model, rather than one aimed purely at prediction, is appropriate. Bidirectional search with BIC is a supported combination (stats::step()), so no custom algorithm is needed. The criterion was fixed before running the analysis.

Question 7. Select predictors (10 points). Apply the chosen search procedure and fixed criterion.

(a) List the predictors in the final selected model.

X1, X4, X10, X13 and X14.

(b) Report the final model formula.

R formula: Y ~ X10 + X4 + X13 + X14 + X1 (listed in the order the variables entered).

Fitted model: $\hat{Y} = 1.9307 + 1.6604 \cdot X1 - 2.6507 \cdot X4 + 7.2586 \cdot X10 - 2.3389 \cdot X13 - 1.8656 \cdot X14$.

(c) Report the chosen criterion's value for the selected model.

BIC = 680.8865 (from BIC(stepwise_BIC)). For comparison, the intercept-only model has BIC = 752.8193.

(d) State the number of selected predictors, excluding the intercept. Briefly describe how you identified the final model along the search path.

5 predictors (intercept excluded).

Search path (step() with k = log(100), trace shown; note that step() labels the column "AIC" even when k = log(N) is used, so these values are BIC up to an additive constant):

```
   Step Df  Deviance Resid. Df Resid. Dev      AIC
1       NA        NA        99   9930.664 464.4264
2 + X10 -1 4706.6253        98   5224.039 404.7959
3  + X4 -1  418.8081        97   4805.231 401.0445
4 + X13 -1  363.4433        96   4441.787 397.7849
5 + X14 -1  362.3603        95   4079.427 393.8800
6  + X1 -1  237.2439        94   3842.183 392.4936
```

Starting from Y ~ 1, the procedure added X10, X4, X13, X14 and X1 in that order. On the equivalent BIC() scale the values were 752.82 → 693.19 → 689.44 → 686.18 → 682.27 → 680.89. At each step it also checked removing the variables already in the model, but no removal ever lowered the criterion. At the final model the best possible addition (X7) would raise the criterion (392.49 → 392.83), as would every removal, so the search stopped. Forward and backward BIC searches (code sections 6.4 and 6.5) reach the same five-predictor model, which supports the result.

Question 8. Examine the selected model (10 points). Refit an ordinary least-squares model using the selected predictors.

(a) Fill in the following table using the R output from the model with selected predictors. If a variable is not selected, leave it empty.

Coefficient	Point estimate	Standard error	t-statistic	p-value
β_1	1.6604	0.6892	2.409	0.017936
β_2				
β_3				
β_4	-2.6507	0.7873	-3.367	0.001103
β_5				
β_6				
β_7				
β_8				

β_9				
β_{10}	7.2586	0.6732	10.782	< 2e-16
β_{11}				
β_{12}				
β_{13}	-2.3389	0.6343	-3.687	0.000379
β_{14}	-1.8656	0.6545	-2.850	0.005366
β_{15}				

(b) Report adjusted R^2.

Adjusted $R^2 = 0.5925$ (Multiple $R^2 = 0.6131$; residual standard error 6.393 on 94 DF; $F = 29.79$ on 5 and 94 DF, $p < 2.2e\text{-}16$).

(c) Compare its adjusted R^2 with that of the full model.

The selected model's adjusted R^2 (0.5925) is slightly higher than the full model's (0.5882), even though it uses 10 fewer predictors. Ordinary R^2 drops from 0.6506 to 0.6131 because R^2 can never increase when variables are removed. Adjusted R^2 penalizes the extra parameters, and by that measure the ten dropped predictors were not worth including.

(d) Discuss whether the selected model is simpler and what information may have been lost by excluding predictors.

Yes, the selected model is much simpler: 5 predictors instead of 15, and every retained coefficient is significant at 5% (largest $p = 0.017936$). Its estimates are close to the full-model estimates, and all five standard errors are slightly smaller.

What may be lost: the ten excluded predictors together explained about 3.75 percentage points of R^2 (RSS rises from 3469.5 to 3842.2). A partial F-test comparing the selected model with the full model gives $F = 0.9023$ on 10 and 84 DF, $p = 0.5349$, so this loss is not statistically significant. However, if any excluded predictor has a small nonzero effect, that effect is now absorbed into the error term and we lose the ability to estimate it. X7 ($p = 0.057$ in the full model, kept by AIC and Cp) and X5 are the most likely candidates. Because the predictors are only weakly correlated (largest $|r| = 0.295$, between X5 and X13; all VIFs below 1.25), omitting them causes little bias in the retained coefficients.

Part 3. LASSO Regression (30 Points)

Question 9. Explain LASSO (5 points). Consider the LASSO objective: min over β_0, β of

$$\frac{1}{2N}\sum_{i=1}^{N}\left(Y_i-\beta_0-\sum_{j=1}^{15}X_{ij}\beta_j\right)^2+\lambda\sum_{j=1}^{15}\left|\beta_j\right|.$$

(a) Explain the role of λ.

λ ≥ 0 is the tuning parameter that controls the strength of the L1 penalty λ·Σ|βⱼ|. It sets the trade-off between fitting the data (the squared-error term) and keeping the coefficients small. When λ = 0, LASSO equals ordinary least squares. As λ increases, all coefficients are shrunk toward zero and more of them become exactly zero, which lowers variance but adds bias. For λ at or above a threshold λ_max (about 6.86 here), every slope coefficient is zero and only the intercept remains.

(b) Explain why LASSO can produce exactly zero coefficient estimates. The intercept is not penalized.

The absolute-value penalty is not differentiable at $\beta_j = 0$. With standardized predictors, the coordinate-wise solution is a soft-thresholding rule: if $z_j = (1/N)\cdot\Sigma\, x_{ij}\cdot(\text{partial residual})_i$, then the LASSO estimate is $\text{sign}(z_j)\cdot\max(|z_j| - \lambda, 0)$. Whenever a predictor's association with the residual, $|z_j|$, is no larger than λ, its estimate is set exactly to 0, not merely made small. Geometrically, the constraint region $\Sigma|\beta_j| \le t$ is a diamond with corners on the coordinate axes, and the RSS contours typically first touch it at a corner, where some coordinates are zero. (Ridge regression's circular region has no corners, so it shrinks coefficients but does not zero them.) The intercept β_0 is not penalized, so it is never shrunk; it just centres the fit.

Question 10. Choose λ by cross-validation (10 points). Use glmnet::cv.glmnet() with 10-fold cross-validation and Gaussian squared-error loss. Use a fixed random seed for reproducibility and standardize predictors using the package's default behavior.

(a) Include the cross-validation plot, plot the cross-validation MSE over λ.

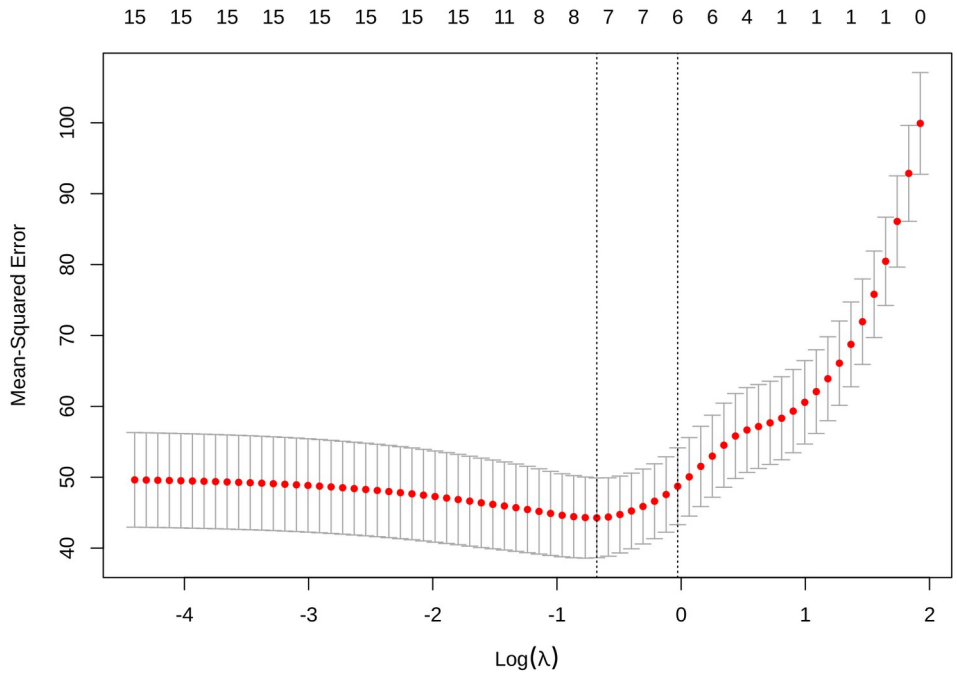


Figure 2. 10-fold cross-validated MSE versus log(λ) (cv.glmnet, seed 6166). Error bars are ±1 SE; the left dashed line marks lambda.min and the right one lambda.1se. Numbers on top give the count of nonzero coefficients.

(b) Report lambda.min, the λ giving the smallest cross-validation MSE.

lambda.min = 0.5070394 (log λ ≈ −0.68). The minimum cross-validation MSE is 44.2725 (SE 5.628), and 8 coefficients are nonzero at this λ.

```
Step 2. Run 10-fold cross-validation

Call:  cv.glmnet(x = X, y = Y, type.measure = "mse", nfolds = 10, alpha = 1,      family =
"gaussian", standardize = TRUE)

Measure: Mean-Squared Error

    Lambda Index Measure    SE Nonzero
min 0.5070    29   44.27 5.628       8
1se 0.9725    22   48.73 5.419       6
```

(c) Explain how the cross-validation criterion selects λ. Use lambda.min throughout the remaining LASSO analysis; do not switch to lambda.1se.

cv.glmnet() computes the LASSO path on a decreasing grid of λ values (from λ_max ≈ 6.86 downward). With set.seed(6166) the 100 observations are randomly split into 10 folds. For each fold k, the model is fitted on the other 9 folds for every λ on the grid, and the squared prediction error is computed on the held-out fold k. Averaging over the 10 folds gives the cross-validation estimate CV(λ) = mean squared prediction error (the red dots), with a standard error from the fold-to-fold spread (the grey bars). lambda.min is the λ that minimizes CV(λ): smaller λ values overfit (higher variance), and larger λ values over-shrink (higher bias). The plot also marks lambda.1se = 0.9725 (6 nonzero coefficients), the largest λ within one SE of the minimum. As instructed, it is not used; lambda.min is used throughout.

Question 11. Fit LASSO and identify variables (10 points). Using the selected λ, obtain the LASSO coefficient estimates.

(a) Report the coefficients.

Enter the LASSO point estimate for each predictor. If LASSO does not select a predictor, enter 0.

Coefficient	**LASSO point estimate**
β_1	1.07944426
β_2	0.00000000
β_3	0.00000000
β_4	-1.92849719
β_5	0.41838736
β_6	0.06818692
β_7	0.89926976
β_8	0.00000000
β_9	0.00000000
β_{10}	6.62840784
β_{11}	0.00000000
β_{12}	0.00000000
β_{13}	-1.62217204
β_{14}	-1.16931949
β_{15}	0.00000000

(b) List all predictors with nonzero coefficients.

X1, X4, X5, X6, X7, X10, X13 and X14 (intercept = 1.69029962, not counted).

(c) State how many predictors LASSO selected. Exclude the intercept from the count.

LASSO selected **8 predictors**; the other 7 (X2, X3, X8, X9, X11, X12, X15) have coefficient 0.

Question 12. Interpret LASSO (5 points). Explain how the penalty affects the fitted coefficients and why some predictors are excluded. Discuss the relationship between the size of λ and model sparsity.

The penalty shrinks every coefficient toward zero relative to OLS. Comparing with the full OLS fit: X10 goes from 7.2157 to 6.6284, X4 from −2.6473 to −1.9285, X13 from −2.6675 to −1.6222, X14 from −1.6059 to −1.1693, X1 from 1.9115 to 1.0794, X7 from 1.2767 to 0.8993, X5 from 1.1445 to 0.4184, and X6 from 0.9065 to 0.0682. Signs are preserved, and the relative shrinkage is largest for the weaker predictors (about 8% for X10 but over 90% for X6).

Predictors are excluded when their contribution to reducing the squared error is not large enough to outweigh the penalty $\lambda|\beta_j|$. By the soft-thresholding rule, their estimate is set exactly to zero. The seven excluded predictors (X2, X3, X8, X9, X11, X12, X15) all had small, clearly non-significant OLS estimates ($|t| < 0.75$). X6 survives only barely (0.068).

Size of λ and sparsity: larger λ means a sparser model. The top axis of Figure 2 shows the number of nonzero coefficients falling from 15 at small λ, to 8 at lambda.min, 6 at lambda.1se, 1 (only X10) for log λ between about 1 and 1.8, and 0 at $\lambda_{max} \approx 6.86$. Smaller λ keeps more predictors, with estimates closer to OLS.

Part 4. Comparison and Discussion (20 Points)

Question 13. Compare selected predictors (10 points). Create a table with one row for each of $X_1, \ldots, X_{15}$ and two columns indicating whether each predictor was selected by (i) classical variable selection and (ii) LASSO. Report (a) the number selected by each method, (b) the predictors selected by both, and (c) those selected by only one method.

Predictor	Classical selection (Yes/No)	LASSO (Yes/No)
X1	Yes	Yes
X2	No	No
X3	No	No
X4	Yes	Yes
X5	No	Yes
X6	No	Yes
X7	No	Yes
X8	No	No
X9	No	No
X10	Yes	Yes
X11	No	No
X12	No	No

X13	Yes	Yes
X14	Yes	Yes
X15	No	No

(a) Classical selection (bidirectional stepwise, BIC) selected **5** predictors; LASSO (lambda.min) selected **8** predictors.

(b) Selected by both: **X1, X4, X10, X13, X14**.

(c) Selected by only one method: **X5, X6, X7** (LASSO only). No predictor was selected by the classical method alone; the classical set is a subset of the LASSO set.

Question 14. Explain differences (5 points). Do the methods select the same predictors? Which selects fewer? Explain at least two reasons the selected sets may differ, considering the criteria, coefficient shrinkage, and correlation among predictors.

No. The methods agree on the five core predictors (X1, X4, X10, X13, X14), but LASSO also keeps X5, X6 and X7. The classical BIC model selects fewer predictors (5 vs 8). Reasons the sets differ:

1. Different criteria/goals. BIC charges $\log(100) \approx 4.61$ per parameter and targets the true sparse model, while lambda.min minimizes estimated prediction error (CV-MSE). Prediction-oriented criteria tend to keep weak predictors that reduce MSE slightly. The same effect is visible within the classical methods: AIC and Cp chose 7 predictors (adding X5 and X7), and adjusted R^2 chose 8 predictors, exactly the LASSO set.

2. Coefficient shrinkage. LASSO estimates are shrunk, so a weak predictor can enter at a small cost with a tiny coefficient (X6 = 0.068, X5 = 0.418). Because lambda.min also shrinks the strong coefficients (e.g., X13 from −2.67 to −1.62), the extra weak terms partly make up for that shrinkage. In subset selection a variable is either out or in at its full unshrunk OLS value, and each addition must pay the full BIC penalty.

3. Correlation among predictors. Here the predictors are only weakly correlated (largest $|r| = 0.295$ between X5 and X13, then 0.258 between X1 and X13; VIFs 1.05–1.24), so collinearity is a minor factor. Still, X5 overlaps with X13 and X14 ($r = 0.295$ and -0.211). Once X13 and X14 are in the model, X5 adds little, and the greedy stepwise search never adds it. LASSO splits the explained variation smoothly among correlated predictors and can keep X5 with a small coefficient. With stronger correlation LASSO tends to pick one of a correlated group somewhat arbitrarily, which can also make the sets differ.

4. Search and randomness. Stepwise selection is a greedy, discrete search, while LASSO follows a continuous path; lambda.min also depends on the random fold assignment (seed 6166).

Question 15. Final discussion (5 points). State which method you prefer for this dataset and justify your choice using model simplicity, the selected predictors, and the results of your analysis. Do not assume that a selected predictor is necessarily a true predictor: the identities of the 5 nonzero population coefficients are unknown to you.

Preferred method: the classical bidirectional stepwise model with BIC (X1, X4, X10, X13, X14).

• **Simplicity:** it uses 5 predictors instead of 8, and every coefficient is significant at 5% ($p \le 0.018$), so the model is easy to interpret.

• **Fit:** its adjusted R^2 (0.5925) is higher than the full model's (0.5882). The ten excluded predictors are jointly non-significant (partial $F = 0.90$, $p = 0.53$).

• **Selected predictors:** its five predictors are exactly the ones both methods agree on, so they have the strongest support. The three LASSO-only predictors have small shrunken coefficients (X6 = 0.068, X5 = 0.418, X7 = 0.899) and were not significant in the full model (p = 0.221, 0.131, 0.057). The LASSO choice is also unstable: the more conservative lambda.1se would keep only 6 predictors.

• **Prediction:** using the same 10 folds as cv.glmnet, the cross-validated MSE of the OLS model with these five predictors is about 42.3, compared with 44.27 for LASSO at lambda.min and 49.9 for the full OLS model. This comparison is somewhat optimistic for the BIC model, because its variables were chosen on the full data, but it shows the simpler model gives up no predictive accuracy.

• **Stability:** forward, backward and bidirectional BIC searches all give the same model.

Caveat: being selected does not prove a predictor is truly nonzero, and the identities of the five true predictors are unknown. X7 is borderline (it is the next variable stepwise BIC would consider, and AIC/Cp keep it), so it could be a real but weak effect. If out-of-sample prediction were the only goal, LASSO would be a reasonable alternative. For identifying a small, interpretable set of predictors, the BIC stepwise model is preferred.

Submission Requirements

Submit only the completed Microsoft Word document (.docx) containing your answers to all project questions.

No additional files are required.

Grading Summary

Component	Points
Part 1. Full regression and diagnostics	20
Part 2. Classical variable selection	30
Part 3. LASSO regression	30
Part 4. Comparison and discussion	20
Total	100